Neonatal Herpes Simplex Infection

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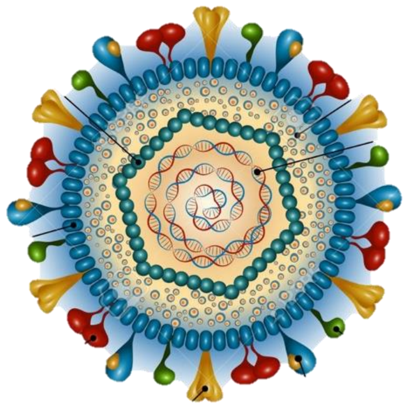
Outlines

- Overview herpes simplex virus
- Epidemiology
- Route of transmission
- Clinical manifestation
- Diagnosis
- Treatment
- Management asymptomatic neonates following vaginal or cesarean delivery to people with active genital herpes leions

Introduction

- Numerous viruses infecting the CNS of neonates, herpes simplex virus (HSV) is among the most severe, with significant mortality and morbidity
- Unlike other viral pathogens, HSV is treatable using a commercially available antiviral drug, acyclovir
- Without treatment, neonatal HSV disease can be fatal and extremely devastating
- It is important for clinicians to recognize signs of neonatal HSV infection
- This can lead to prompt diagnosis and treatment, leading to a better prognosis

Herpes Simplex Virus



- HSVs are large, enveloped, double-stranded DNA viruses
- Family Herpesviridae and, subfamily Alphaherpesviridae
- Two distinct HSV types exist: HSV-1 and HSV-2
- HSV-1 traditionally involves face and skin above the waist; however, increasing number of genital herpes attributable to HSV-1
- HSV-2 usually involves genitalia and skin below waist in sexually active adolescents and adults
- Both HSV-1 and HSV-2 cause herpetic disease in neonates
- HSV-1 and HSV-2 establish latency in sensory neurons following primary infection, with periodic reactivation causing recurrent symptomatic disease or asymptomatic viral shedding
- Genital HSV-2 infection is more likely to recur than is genital HSV-1 infection

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Epidemiology

- The incidence of neonatal HSV infection is between 1 in 3,000 and 1 in 20,000 live births
- The incidence of neonatal HSV infection in the United States has increased over the past 2 decades to approximately 1 in 2000 live births
- More than three quarters of infants who contract HSV infection are born to a person wit no history or clinical findings suggestive of genital HSV infection during or preceding pregnancy
- Therefore, lack of history of genital HSV infection in the birthing parent does not preclude a diagnosis of neonatal HSV disease

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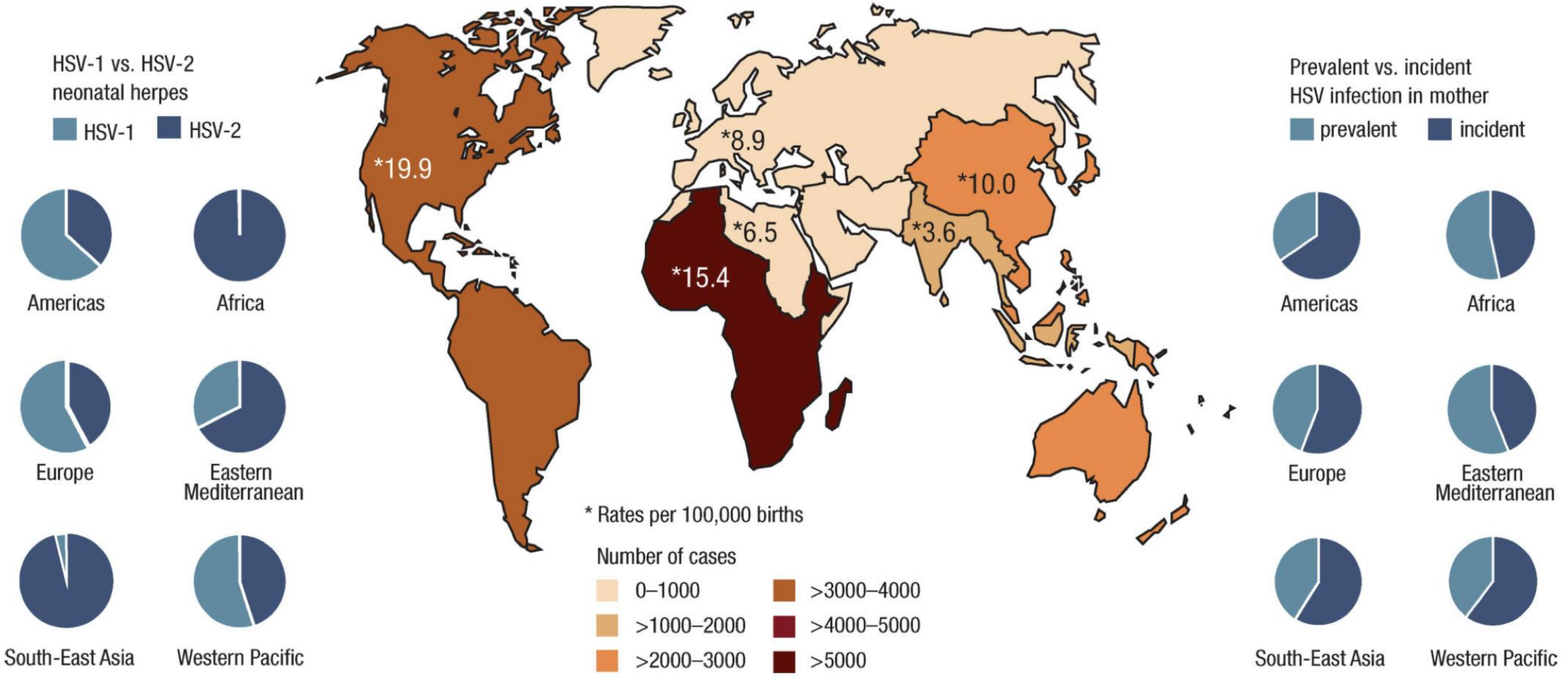
Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Estimates of the annual number of cases, and rate per 100,000 births, of neonatal herpes during 2010–2015



Looker KJ, et al, Lancet Glob Health. 2017 Mar;5(3):e300-e309.

Transmission

01

Intrauterine:
5%

02

Peripartum:
85%

03

Postpartum:
10%

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Transmission

- HSV is transmitted to a neonate most often during birth through an infected genital tract but can be caused by an ascending infection through ruptured or apparently intact amniotic membranes
- Other less common sources of neonatal infection
  - Postnatal transmission from a parent, sibling, or other caregiver, most often from a nongenital infection (eg, mouth or hands), transmission from the mouth of a religious circumciser ("mohel") to the infant penis during ritual Jewish circumcisions that include direct orogenital suction (metzitzah b'peh)
  - Intrauterine infection causing congenital malformations

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Transmission

- Risk factors that increase the likelihood of HSV transmission from a pregnant woman who is shedding HSV genitally to her infant include:
  - 1. Type of maternal infection (primary vs recurrent)
  - 2. Maternal antibody status
  - 3. Longer duration of rupture of membranes
  - 4. Integrity of mucocutaneous barriers (using fetal scalp probe, incisions, etc)
  - 5. Mode of delivery (cesarean vs vaginal delivery)

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Transmission

- Infants of mothers with primary genital HSV infection at delivery have a much higher risk (25 – 60%) than those of mothers with recurrent infection (< 2%)
- This increased risk is because primary infection
  - → Lower maternal HSV IgG transfer (no time to generate, less effective binding)
  - → Higher, prolonged viral shedding vs recurrent infection

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Congenital HSV Infection

- In utero HSV transmission, is the least common route of MTCT, with estimated transmission rate of 1 in 300,000 deliveries in the United States
- It presents with distinct symptoms triad of cutaneous, neurologic, and ocular manifestations already present at the time of birth
- Intrauterine HSV infection occur with both primary and recurrent maternal HSV infections, although the risk from a recurrent infection is less.

Table 1
Clinical manifestations of congenital HSV infection

Cutaneous	Neurologic	Ocular
Active herpetic lesions	Microcephaly	Optic atrophy
Scarring	Intracranial calcifications	Chorioretinitis
Aplasia cutis	Hydranencephaly	Microphthalmia
Hyperpigmentation or hypopigmentation	—	—

James Sh, et al.Infect Dis Clin North Am. 2015 Sep;29(3):391-400.

Case Report Intrauterine HSV Infection

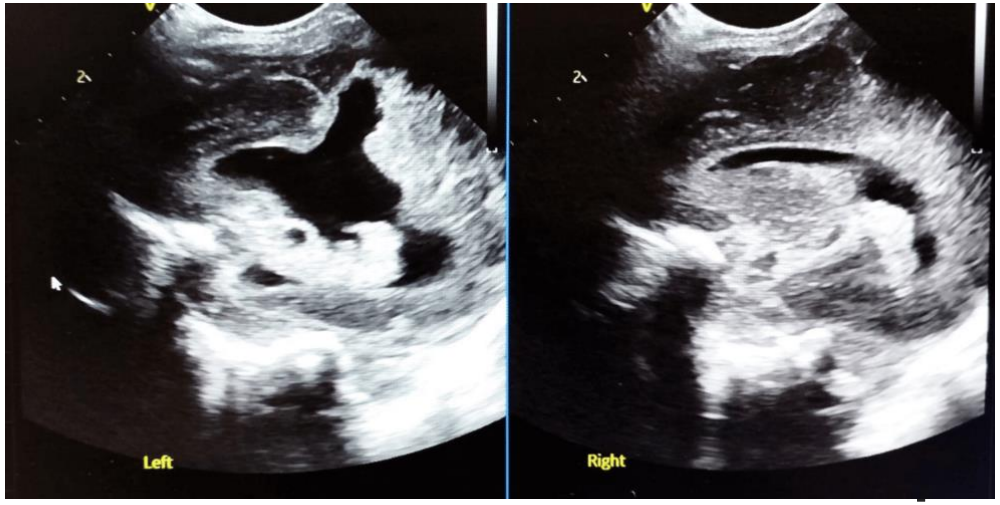
Maternal history: 42-year-old gravida 4 para 3

- No symptoms of genital HSV before or during pregnancy
- First and second trimester ultrasounds: normal
- Prenatal screening negative (HIV, syphilis, hepatitis B/C, TORCH except immune to rubella & toxo)

Pregnancy complications

- **PROM at 24 weeks** → ultrasound showed:
  - Severe cerebral malformations
  - Anasarca, pleural/pericardial effusion
  - Cardiomegaly, hepatomegaly with intrahepatic calcifications

•Infectious testing (CMV, parvovirus, Zika) negative
•Amniocentesis: normal microarray



Oliveira Í, et al. Cureus 17(1): e78173.

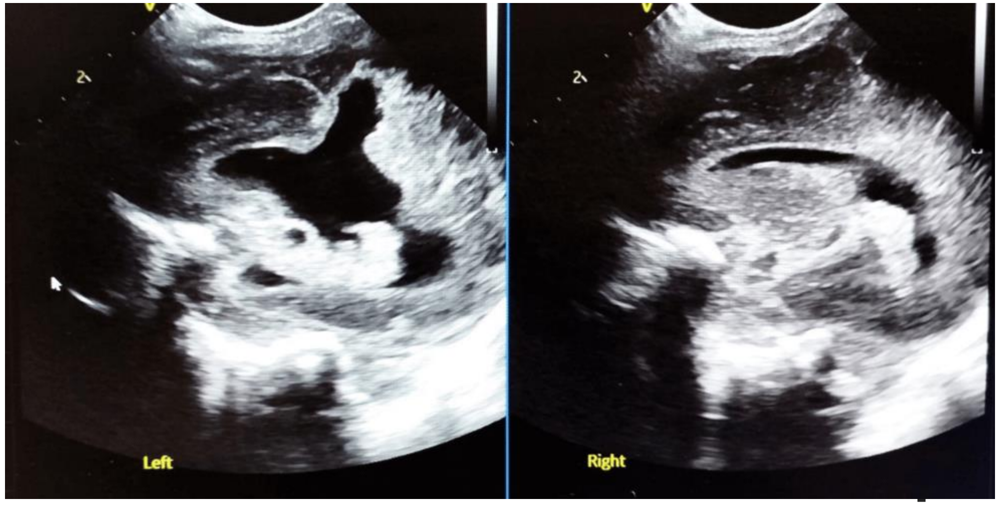
Case Report Intrauterine HSV Infection

Delivery: Emergency C/S at GA 31+3 weeks

- Female infant, 1130 g, HC 23 cm (<3rd percentile)
- PE: diffuse skin erosions (abdomen, forearm, lower limbs)
- Abdominal US: hepatic/suprarenal/renal calcifications
- Cranial US: severe CNS malformations

Clinical progression

- Day 4: grouped vesicular lesions appeared
- HSV suspected → IV acyclovir (20 mg/kg q8h) started
- PCR skin swab: HSV-2 positive (high viral load ~96 million copies/mL)
- Serum: HSV-1/2 IgG positive, IgM negative
- Died on day 5 despite supportive care



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Clinical Manifestations

In newborn infants with herpes simplex virus (HSV), infection can manifest as

25% — **Disseminated disease involving multiple organs**, most prominently live and lungs, and also involving the central nervous system (CNS)

30% — **Localized CNS disease**, with or without skin, eye, or mouth involvement (CNS disease)

45% — **Disease localized to the skin, eyes, and/or mouth (SEM disease)**

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Disseminated HSV Disease

- Severe hepatitis, disseminated intravascular coagulation, pneumonitis, possibly CNS involvement (found in 60%–75% of cases)
- Interestingly, 40% of disseminated HSV disease do not develop skin findings
- Approximately two thirds of neonates with disseminated or CNS disease have skin vesicles, but these lesions may not be present at the time of onset of symptoms
- The mean age at presentation of disseminated infection is approximately 11 days after birth

Disseminated infection should be considered in neonates with sepsis syndrome with negative bacteriologic culture results, severe liver dysfunction, consumptive coagulopathy, or suspected viral pneumonia, especially hemorrhagic pneumonia

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Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Neonatal HSV CNS disease

- Can present as seizures (focal or generalized), lethargy, poor feeding, irritability, tremors, temperature instability, and bulging fontanelle
- The mean age at presentation for CNS disease is approximately 16 days after birth
- Approximately 60% to 70% of infants with CNS disease will also have skin manifestations at some point in the disease course
- Mortality is usually due to devastating brain destruction and atrophy, causing neurologic and autonomic dysfunction

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

SEM disease

- SEM disease is associated with the best outcomes, with virtually no mortality and with morbidity associated solely with cutaneous recurrences but no neurologic sequelae
- In addition, infants with SEM disease are most likely to have skin lesions (in >80% of patients), which facilitates diagnosis and allows prompt initiation of antiviral treatment before the disease progresses to involve other organs
- Presenting signs and symptoms of SEM disease include skin vesicles, fever, lethargy, and conjunctivitis
- Mean age at presentation for SEM disease is approximately 12 days after birth

Neonatal Herpes Simplex Virus Infection (SEM disease)

Infant: Preterm girl, GA 32 week, ND

Maternal history: 37-year-old mother

- **Premature rupture of membranes (PROM) 12 days before delivery**
- No genital or placental lesions observed, no known prior HSV-1 infection
- **Postpartum endocervical PCR: positive HSV 1** → suggests **primary infection late pregnancy**

Neonatal findings:

- Vesicular lesions at torso, periumbilical region, pharyngeal/nasal/conjunctival mucosa
- No CNS, pulmonary, or hepatic involvement (normal chest X-ray, LP, LFTs)
- PCR positive for HSV-1 from skin, conjunctivae, pharynx, and blood

Mantecón L, et al. N Engl J Med. 2023 Mar 30;388(13):1209.

Neonatal Herpes Simplex Virus Infection (SEM disease)

Infant: Preterm girl, GA 32 week, ND

Diagnosis:

- **Neonatal HSV-1 infection confined to skin, eyes, mucosa (SEM disease)**
- Likely **ascending intrauterine infection** after prolonged ROM, associated with preterm labor

Management:

- Treated with **acyclovir**

Outcome:

- At 2 years → no recurrences, normal development

Mantecón L, et al. N Engl J Med. 2023 Mar 30;388(13):1209.

Clinical Manifestations

- Both HSV type 1(HSV-1) and type 2 (HSV-2) can cause any of these manifestations of neonatal HSV disease
- In the absence of characteristic skin lesions, the diagnosis of neonatal HSV infection is challenging and requires a high index of suspicion
- More than 80% of neonates with SEM disease have skin vesicles; those without vesicles have infection limited to the eyes and/or oral mucosa
- HSV should be considered as a causative agent in neonates with fever (especially within the first 3 weeks of life), hypothermia, a vesicular rash, or abnormal cerebrospinal fluid (CSF) findings (especially in the presence of seizures or during a time of year when enteroviruses are not circulating in the community)

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Clinical Manifestations

- Although asymptomatic HSV infection is common in older children, it rarely if ever, occurs in neonates
- Recurrent skin lesions are common in surviving infants, occurring in approximately 50% of survivors, often within 1 to 2 weeks of completing the initial treatment course of parenteral acyclovir or the longer-term suppressive course of oral acyclovir

Clinical Manifestations

- Initial signs of HSV infection can occur anytime between birth and approximately 6 weeks of age, although almost all infected infants develop clinical disease within the first month of life
- Infants with disseminated disease and SEM disease have an earlier age of onset, typically presenting between the first and second weeks of life
- Infants with CNS disease usually present with illness between the second and third weeks of life

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Diagnosis

- A nucleic acid amplification test (NAAT) usually can detect HSV DNA in CSF and is the diagnostic method of choice for CNS HSV involvement
- A NAAT of CSF can yield negative results in cases of HSE, especially early in the disease course
- In cases which HSV CNS disease is expected but repeated CSF NAAT results are negative, histologic examination and viral culture of a brain tissue biopsy is most definitive method of confirming the diagnosis of HSE
- Detection of intrathecal antibody against HSV also can assist in the diagnosis but may not be specific because of alterations in the blood-brain barrier
- Viral cultures of CSF from a patient with HSE usually are negative

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Diagnosis

- HSV grows readily in traditional cell culture
- Cytopathogenic effects typical of HSV infection usually are observed 1 to 3 days after inoculation
- Methods of culture confirmation include fluorescent antibody staining, enzyme immunoassays (EIAs), and monolayer culture with typing
- Cultures that are negative on day 5 likely will remain negative
- A viral isolate by culture is required if phenotypic antiviral susceptibility studies are to be performed

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Diagnosis of neonatal HSV infection

All of the following specimens should be obtained for each patient

1. Swab specimens from the conjunctivae, mouth, nasopharynx, and anus (surface specimens) for HSV culture (if available) or NAAT (can use a separate swab for each site, or a single swab starting with the conjunctivae)
2. Specimens of skin vesicles for HSV culture (if available) or NAAT
3. CSF sample for HSV NAAT
4. Whole blood sample for HSV NAAT
5. Whole blood sample for measuring alanine aminotransferase (ALT)

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Diagnosis of neonatal HSV infection

- Any neonatal HSV disease (disseminated, CNS, SEM) can have associated viremia, positive whole blood NAAT does not define as having disseminated HSV and, therefore, should not be used to determine extent of disease and duration of treatment
- Likewise, no data exist to support use of serial blood NAAT to monitor response to therapy
- Rapid diagnostic techniques, such as direct fluorescent antibody staining of vesicle scrapings or ELA detection of HSV antigens are as specific but less sensitive than culture

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Diagnosis of neonatal HSV infection

- Type-specific antibodies to HSV develop during the first several weeks after infection and persist indefinitely
- Approximately 20% of patients with a first episode of HSV-2 seroconvert for type-specific IgG antibody by 10 days, and the median time to seroconversion is 21 days with a type-specific ELISA; more than 95% of people seroconvert by 12 weeks following infection
- Serologic testing is not useful in neonates
- IgM testing for HSV-1 or HSV-2 is not useful because of lack of a reliable commercially available IgM assay

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Diagnosis of neonatal HSV infection

- In the past, the presence of RBC in CSF was suggestive of HSV CNS infection, likely as a result of relatively advanced disease due to diagnostic limitations
- However, more advanced imaging and diagnostic capabilities, hemorrhagic HSV encephalitis is less commonly seen now, and as such, most HSV CNS CSF indices do not have significant numbers of red blood cells
- Performance of whole blood PCR adds to the other diagnostic tools (surface and CSF cultures and CSF PCR), but should not be used as the sole test for ruling in or ruling out neonatal HSV infection
- Furthermore, viremia and DNAemia can occur in any of the 3 types of neonatal HSV disease, so a positive whole blood PCR simply rules in neonatal HSV infection but does not assist in disease classification

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Diagnosis of neonatal HSV infection

- All infants with HSV disease, regardless of classification, need to have an ophthalmologic examination to look for ocular involvement such as uveitis, conjunctivitis, and keratitis
- Infected neonates with any extent of disease manifestations should undergo neuroimaging studies (MRI preferably, but head CT or ultrasonography are acceptable) to establish baseline brain anatomy
- Later findings can include brain abscesses (particularly in the temporal lobe) or severe encephalomalacia

Infection Classification by Genital HSV Viral Type and Type-Specific Serologic Test Results From the Birthing Parent

Classification of Infection in Birthing Parent	NAAT/Culture From Genital Lesion	HSV-1 and HSV-2 IgG Type-Specific Antibody Status of Birthing Parent
Documented first-episode primary infection	Positive, either virus	Both negative
Documented first-episode nonprimary infection	Positive for HSV-1	Positive for HSV-2 **AND** negative for HSV-1
	Positive for HSV-2	Positive for HSV-1 **AND** negative for HSV-2
Assumed first-episode (primary or nonprimary) infection	Positive for HSV-1 **OR** HSV-2	Not available
	Negative **OR** not available[b]	Negative for HSV-1 and/or HSV-2, **OR** not available
Recurrent infection	Positive for HSV-1	Positive for HSV-1
	Positive for HSV-2	Positive for HSV-2

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Treatment

- Parenteral acyclovir is the treatment for neonatal HSV infections
  - The dosage of acyclovir is 60 mg/kg per day in 3 divided doses (20 mg/kg/dose)
  - Administered intravenously for 14 days in SEM disease and for a minimum of 21 days in CNS disease or disseminated disease
- All infants with CNS involvement should have a repeat lumbar puncture performed near the end of therapy to document that the CSF is negative for HSV DNA on NAAT
- In the unlikely event that the NAAT result remains positive near the end of a 21-day treatment course, intravenous acyclovir should be administered for another week, with repeat CSF NAAT performed near the end of the extended treatment period and another week of parenteral therapy if the result remains positive

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Treatment

- Parenteral antiviral therapy should not be stopped until the CSF NAAT result for HSV DNA is negative
- Infants surviving neonatal HSV disease of any classification (disseminated, CNS, or SEM) should receive oral acyclovir suppression at 300 mg/m^2/dose, administered 3 times daily for 6 months after the completion of parenteral therapy for acute disease; the dose should be adjusted monthly to account for growth

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Treatment

Generic (Trade Name)	Indication	Route	Age	Usually Recommended Dosage
Acyclovir[b,c,d,e] (Zovirax)	Neonatal herpes simplex virus (HSV) infection	IV	Birth to ≤4 mo	Treatment dosing: 60 mg/kg per day, in 3 divided doses for 14 days (SEM disease) or 21 days (CNS or Disseminated disease) (durations >21 days are necessary if CSF PCR remains positive near end of treatment course)
		Oral	2 wk to 8 mo	Oral suppressive dosing following completion of IV treatment; dosing: 300 mg/m^2, 3 times per day for 6 mo

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Treatment

- Absolute neutrophil counts should be assessed at 2 and 4 weeks after initiating suppressive acyclovir therapy and then monthly during the treatment period
- Longer durations or higher doses of antiviral suppression do not further improve neurodevelopmental outcomes
- Valacyclovir has not been studied for longer than 5 days in young infants, so it should not be used routinely for antiviral suppression in this age group

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Treatment

- Topical ophthalmic drug (1%trifluridine or 0.15% ganciclovir), in addition to parenteral antiviral therapy
  - Indicated in infants with ocular involvement attributable to HSV infection
  - Including a positive virologic test result from a conjunctival swab sample
  - An ophthalmologist should be involved in the management and treatment of acute neonatal ocular HSV disease

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Infection and Prevention Control Measures

Neonate with HSV infection:

- Neonates with HSV infection should be hospitalized and managed with contact precautions if mucocutaneous lesions are present

Neonate exposed to HSV during delivery:

- An infant born to a person with active genital HSV lesions should be managed with contact precautions during the incubation period
- The risk of HSV infection in an infant born to a person with a history of recurrent genital herpes who has no genital lesions at delivery is low so special precautions are not necessary

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Infection and Prevention Control Measures

Postpartum Person With HSV Infection:

- People with active HSV lesions should be instructed about the importance of careful hand hygiene before and after caring for their infants
- A parent or caregiver with herpes labialis or stomatitis should wear a disposable surgical mask when touching the newborn infant until the lesions have crusted
- The caregiver should not kiss or nuzzle the newborn infant until lesions have resolved
- Herpetic lesions on other skin sites should be covered

Breastfeeding

- Breastfeeding is acceptable if no lesions are present on the breasts and if active lesions elsewhere on the caregiver are covered

Outcomes

- Until recently, improvement in morbidity outcomes after antiviral treatment was less dramatic than mortality for neonates with disseminated disease or CNS disease
- Oral acyclovir suppressive therapy has significantly improved the neurologic outcomes of infants with CNS involvement
- Without treatment, 50% of neonates who survived disseminated HSV disease were developing normally at 1 year of age
- With the use of higher-dose acyclovir for 21 days the number of infants developing normally at 1 year of age after disseminated HSV disease has increased to 83%

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Outcomes

- Similarly, for CNS HSV disease, 33% of patients demonstrate normal neurologic development at 1 year of age after 10 days of lower-dose acyclovir therapy, compared with 31% of children treated with higher-dose acyclovir for 21 days
- However, with concomitant use of oral acyclovir therapy for 6 months, this percentage of infants with normal neurodevelopment at 1 year increases to 69%
- Morbidity of SEM disease also has dramatically improved since the introduction of antiviral treatment
- In the pre-antiviral era, 38% of patients with SEM disease were developing normally at 1 year of age, but with antiviral therapy, this risk is eliminated completely (due to SEM disease not progressing to CNS or disseminated disease)

Kabani, N., et al. *NeoReviews*, *19*(2), e89-e96.

Mortality among patients with known outcomes after 12 months of age

Table 2
Mortality and neurodevelopmental impairment in neonatal HSV infection

Treatment	Mortality at 1 y (%)		Neurodevelopmental Impairment (%)	
	Disseminated	CNS	Disseminated	CNS
Preantiviral era[32]	85	50	50	67
Vidarabine[41]	50	14	42	43
Acyclovir (30 mg/kg/d)[41]	61	14	40	71
Acyclovir (60 mg/kg/d)[34]	29	4	17	69
Acyclovir (60 mg/kg/d) followed by 6 mo of oral suppressive acyclovir therapy (300 mg/m^2/dose administered 3 times per day)[42]	Not applicable	Not applicable		31[a]

[a] Includes infants with CNS disease as well as infants with CNS involvement in the setting of disseminated disease.

James Sh, et al.Infect Dis Clin North Am. 2015 Sep;29(3):391-400.

Prevention of Neonatal Infection

During Pregnancy

- The American College of Obstetricians and Gynecologists recommends that people with a primary or nonprimary first-episode outbreak in pregnancy as well as those with a clinical history of genital herpes, should be offered suppressive antiviral therapy at or beyond 36 weeks of gestation
- However, cases of neonatal HSV disease have occurred among infants born to people who received such antiviral prophylaxis
- First-episode primary and nonprimary genital infections can result in shedding of HSV in the genital tract for several weeks even after lesions resolve

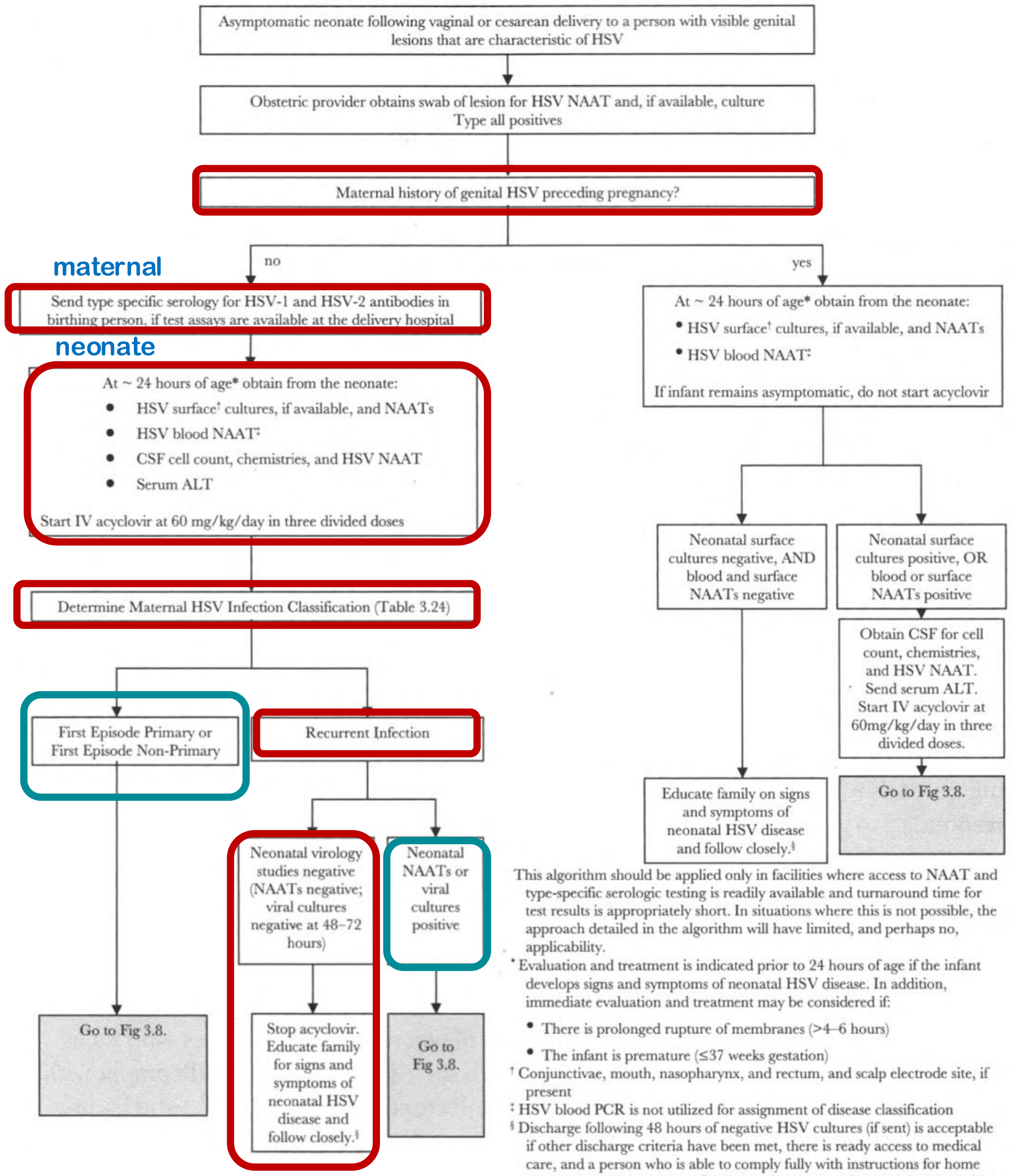
Prevention of Neonatal Infection

Care of Newborn Infants Whose Birthing Parents Has Active Genital Lesion at Delivery

- Neonates exposed to active HSV genital lesions at delivery should not have delayed bathing
- The risk of transmitting HSV to the newborn infant during delivery is influenced directly by the classification of HSV infection of the birthing individual
- Those with primary genital HSV infections who are shedding HSV at delivery are 10 to 30 times more likely to transmit the virus to their newborn infants, compared with those with a recurrent genital infection

The Evaluation Of Asymptomatic Neonates Following Vaginal Or Cesarean Delivery To People With Active Genital Herpes Lesions

*Evaluation and treatment is indicated prior to 24 hours of age if the infant develops signs and symptoms of neonatal HSV disease
In addition, immediate evaluation and treatment may be considered if:
• There is prolonged rupture of membranes (>4- 6 hours)
* The infant is premature (GA≤37 weeks)



This algorithm should be applied only in facilities where access to NAAT and type-specific serologic testing is readily available and turnaround time for test results is appropriately short. In situations where this is not possible, the approach detailed in the algorithm will have limited, and perhaps no, applicability.

* Evaluation and treatment is indicated prior to 24 hours of age if the infant develops signs and symptoms of neonatal HSV disease. In addition, immediate evaluation and treatment may be considered if:
  - There is prolonged rupture of membranes (>4–6 hours)
  - The infant is premature (≤37 weeks gestation)

† Conjunctivae, mouth, nasopharynx, and rectum, and scalp electrode site, if present

‡ HSV blood PCR is not utilized for assignment of disease classification

§ Discharge following 48 hours of negative HSV cultures (if sent) is acceptable if other discharge criteria have been met, there is ready access to medical care, and a person who is able to comply fully with instructions for home observation will be present. Discharge does not need to be delayed awaiting NAAT results if all other above criteria have been met and a clear plan to follow-up on all virology results is in place. If any of these conditions is not met, the infant should be observed in the hospital until all HSV testing is finalized as negative or HSV cultures are negative for 96 hours after being set up in cell culture, whichever is shorter.

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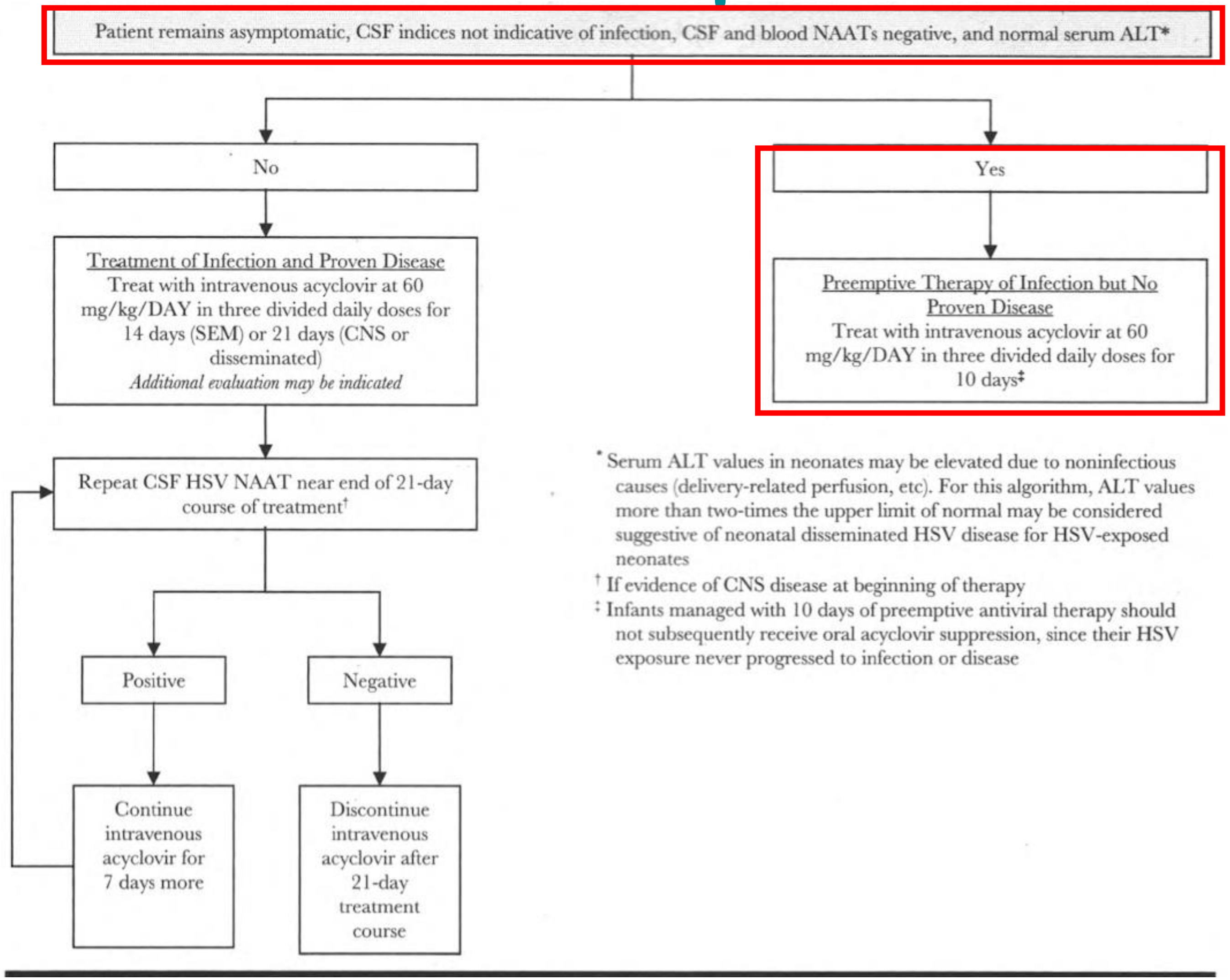

* Serum ALT values in neonates may be elevated due to noninfectious causes (delivery-related perfusion, etc). For this algorithm, ALT values more than two-times the upper limit of normal may be considered suggestive of neonatal disseminated HSV disease for HSV-exposed neonates

† If evidence of CNS disease at beginning of therapy

‡ Infants managed with 10 days of preemptive antiviral therapy should not subsequently receive oral acyclovir suppression, since their HSV exposure never progressed to infection or disease

Preemptive treatment 10 days, without supperssive therapy

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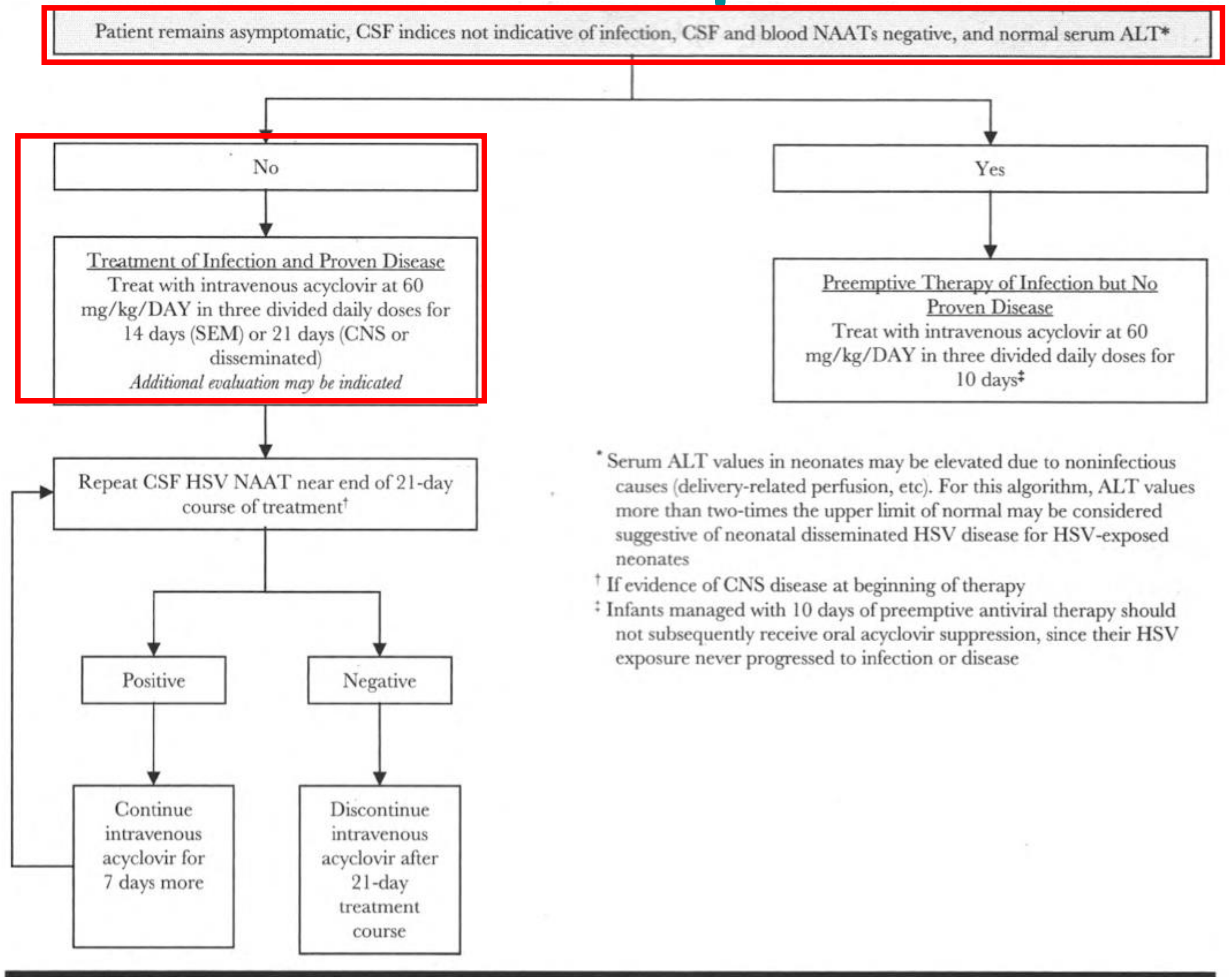

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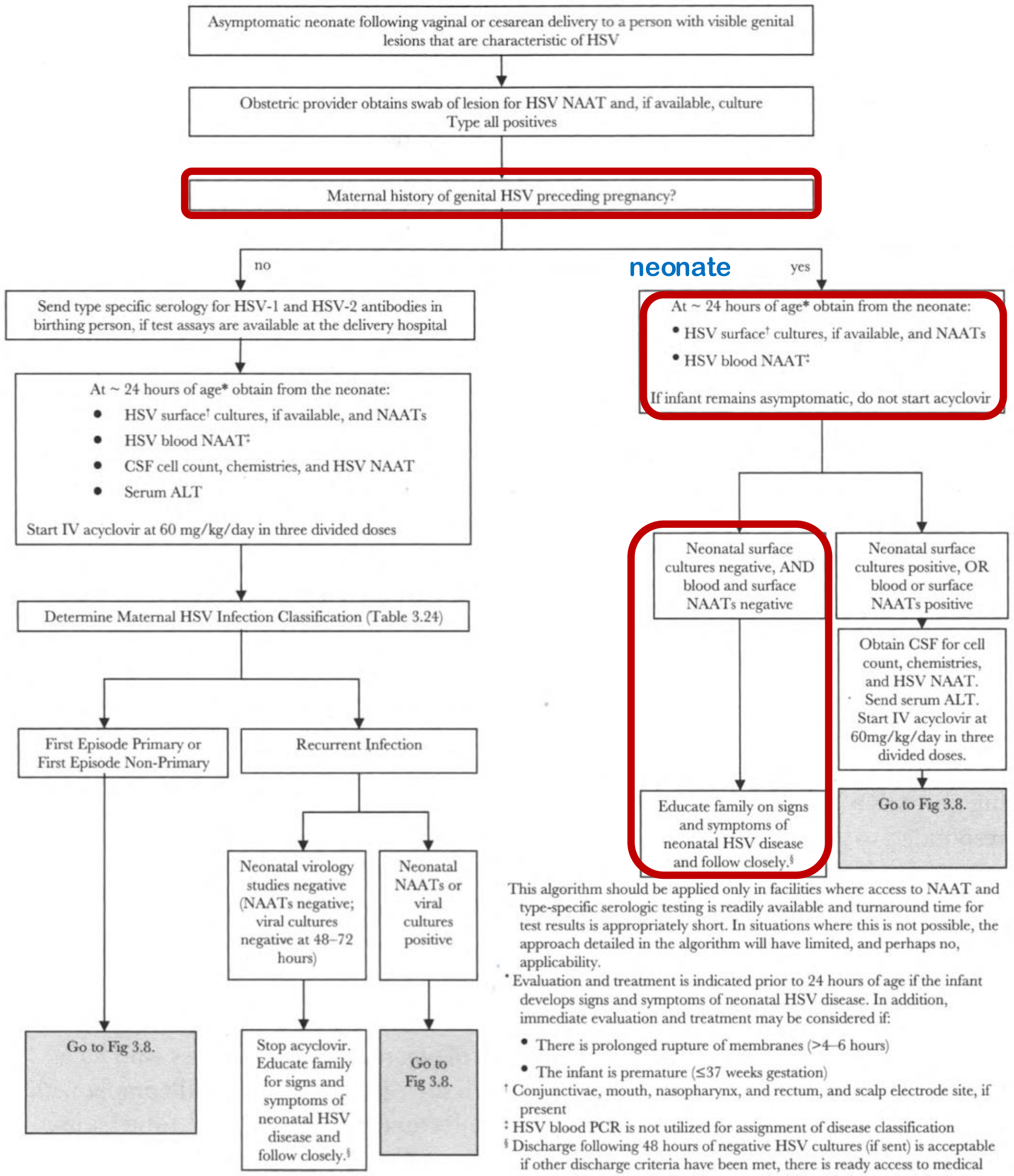

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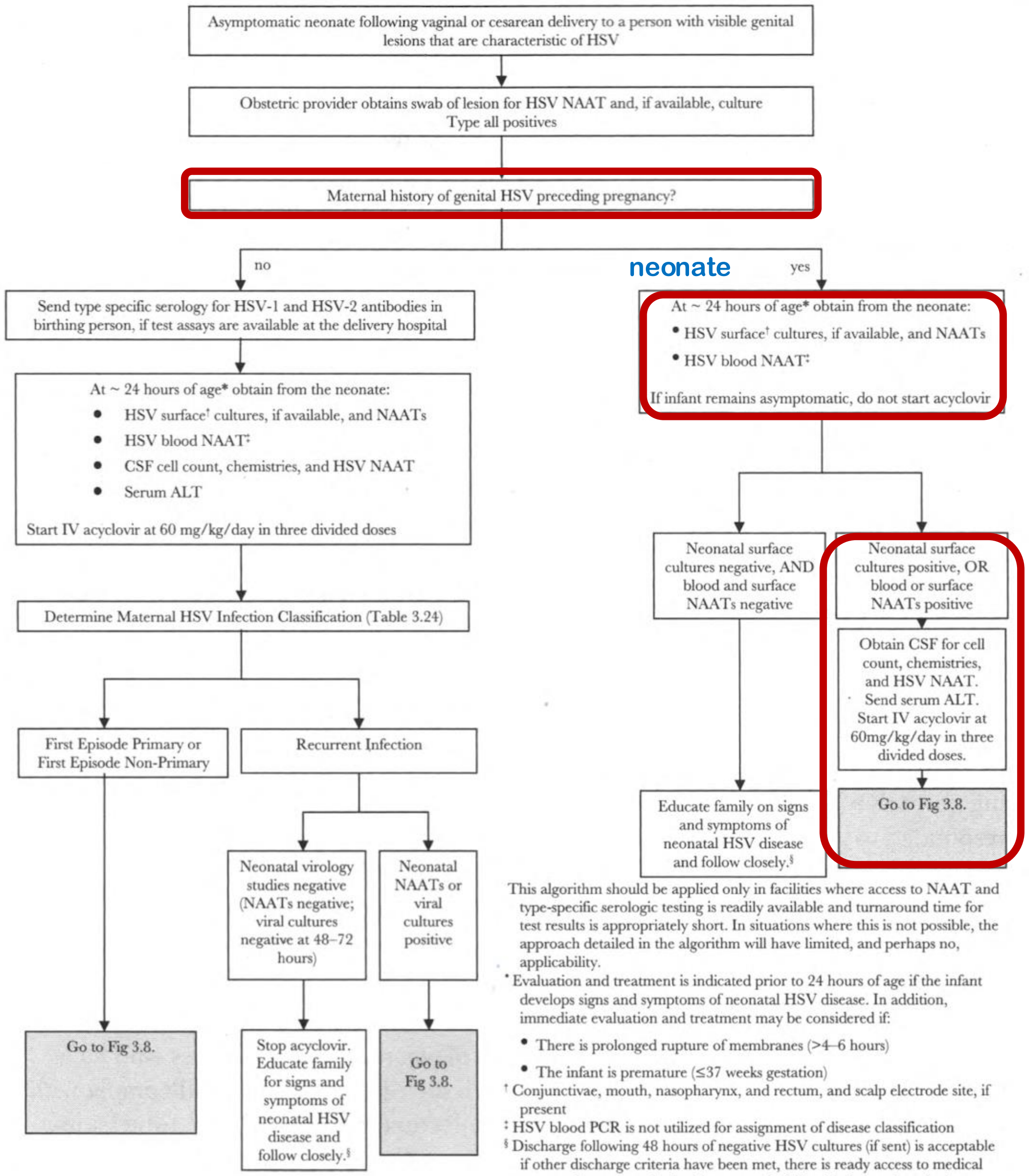

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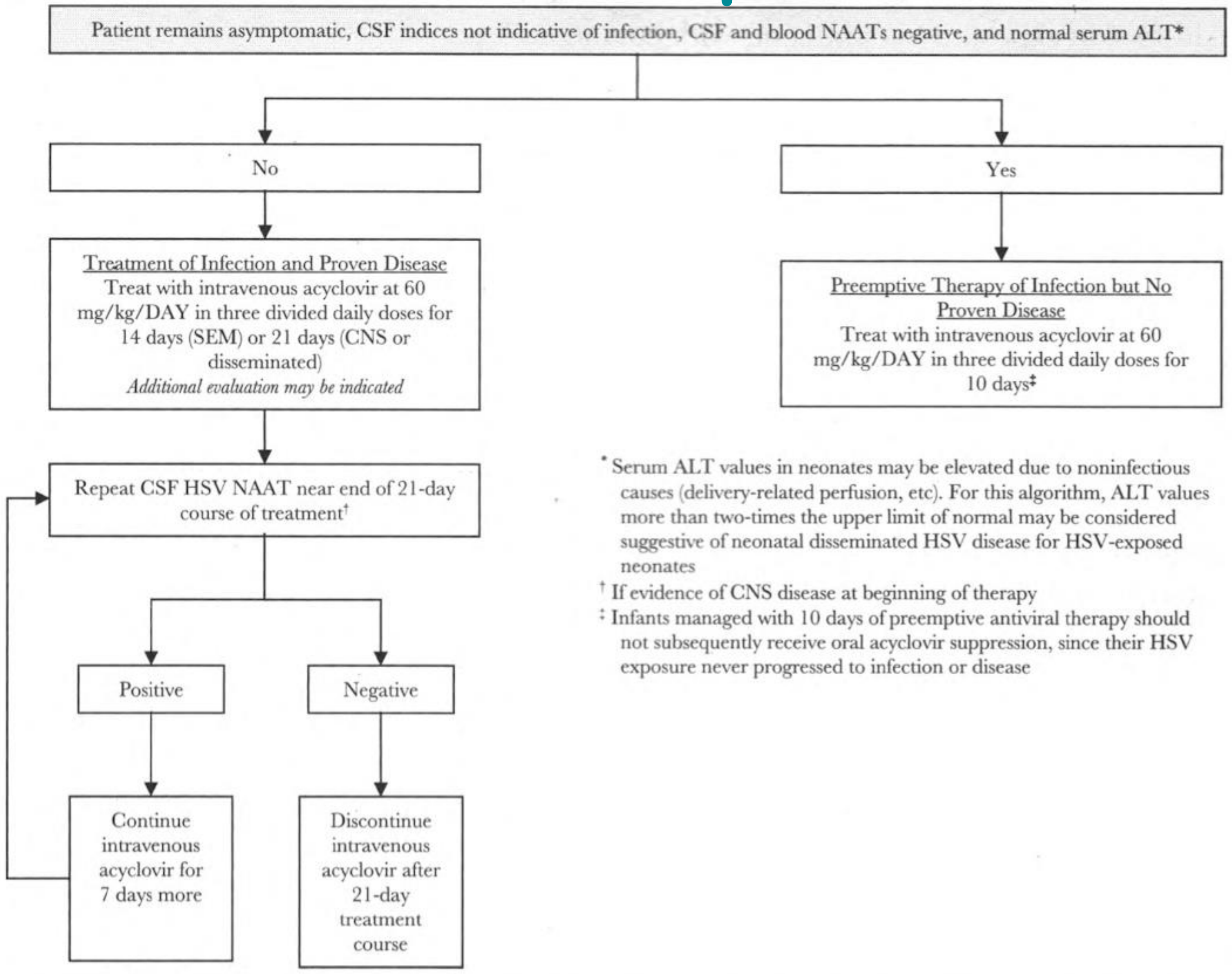

* Serum ALT values in neonates may be elevated due to noninfectious causes (delivery-related perfusion, etc). For this algorithm, ALT values more than two-times the upper limit of normal may be considered suggestive of neonatal disseminated HSV disease for HSV-exposed neonates

† If evidence of CNS disease at beginning of therapy

‡ Infants managed with 10 days of preemptive antiviral therapy should not subsequently receive oral acyclovir suppression, since their HSV exposure never progressed to infection or disease

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Prevention of Neonatal Infection

Care of Newborn infants Whose Birthing Parent Has a History of Genital Herpes But No Active Genital Lesions at Delivery

- Most infants whose birthing parent has a history of genital herpes but no genital lesions at delivery should be observed for signs of infection (eg, vesicular lesions of the skin, respiratory distress, seizures, or signs of sepsis)
- But should not have specimens for surface cultures for HSV obtained at 12 to 24 hours of life and should not receive empiric parenteral acyclovir

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- The exception are infants born to persons with documented first-episode primary or nonprimary infections during the third trimester but who have no genital lesions at delivery
- These infants should have HSV surface cultures, if available, and NAATs and HSV blood NAAT obtained at 24 hours of age, regardless of mode of delivery or birthing parent's receipt of antiviral suppressive therapy

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- The rationale for this recommendation is that the American College of Obstetricians and Gynecologists states that cesarean delivery may be offered to people with documented first-episode infection during the third trimester because of the possibility of prolonged viral shedding
- Education of parents and caregivers about the signs and symptoms of neonatal HSV infection during the first 6 weeks of life is prudent

Thank you